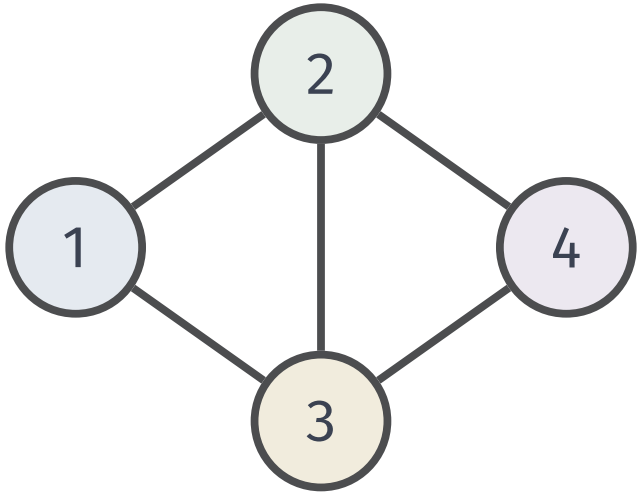
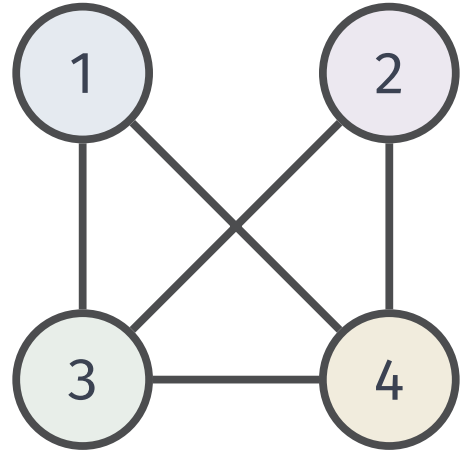


(a)

$\rho(1) = 1$
$\rho(2) = 3$
$\rho(3) = 4$
$\rho(4) = 2$

(b)



(c)